Answer Key 7

Q	Answer	Q	Answer	Q	Answer
1)	4	61)	4	121)	2
2)	3	62)	3	122)	3
3)	1	63)	2	123)	1
4)	4	64)	1	124)	1
5)	2	65)	4	125)	1
6)	1	66)	2	126)	1
7)	3	67)	1	127)	2
8)	3	68)	4	128)	3
9)	1	69)	2	129)	4
10)	2	70)	2	130)	4
11)	1	71)	3	131)	2
12)	2	72)	3	132)	4
13)	4	73)	1	133)	2
14)	2	74)	2	134)	3
15)	2	75)	4	135)	2
16)	1	76)	3	136)	2
17)	2	77)	2	137)	4
18)	1	78)	3	138)	4
19)	3	79)	3	139)	1
20)	1	80)	2	140)	4
21)	3	81)	3	141)	3
22)	2	82)	1	142)	1
23)	3	83)	3	143)	3
24)	2	84)	2	144)	1
25)	3	85)	4	145)	3
26)	1	86)	3	146)	4
27)	1	87)	3	147)	3
28)	3	88)	1	148)	4
29)	2	89)	4	149)	4
30)	3	90)	2	150)	3
31)	3	91)	1	151)	1
32)	4	92)	1	152)	1
33)	4	93)	1	153)	1
34)	4	94)	2	154)	4
35)	3	95)	4	155)	3
36)	4	96)	3	156)	2
37)	2	97)	3	157)	1
38)	3	98)	2	158)	4
39)	4	99)	1	159)	4
40)	3	100)	2	160)	4
41)	1	101)	3	161)	4
42)	1	102)	4	162)	2
43)	3	103)	2	163)	3
44)	4	104)	1	164)	3
45)	1	105)	4	165)	4
46)	1	106)	2	166)	2
47)	1	107)	2	167)	1
48)	3	108)	3	168)	2
49)	1	109)	4	169)	1
50)	4	110)	1	170)	4
51)	2	111)	4	171)	2
52)	1	112)	3	172)	1
53)	3	113)	2	173)	1
54)	4	114)	1	174)	2
55)	2	115)	3	175)	2
56)	2	116)	2	176)	2
57)	3	117)	4	177)	2
58)	4	118)	1	178)	2
59)	1	119)	3	179)	2
60)	3	120)	4	180)	4

Q1) **Explanation:**

Hint: $\Delta U_{\text{cycle}} = 0$

Step: Find the change in the internal energy of the system during process $3 \to 1.$
For a closed cycle (which the system undergoes), the net change in internal energy over the entire cycle is zero i.e., $\Delta U = 0$
So, the total heat added to the system is equal to the work done by the system over the entire cycle i.e.,
$Q_{\text{total}} = W_{\text{total}} \quad \ldots (1)$
The total heat added is the sum of the heat exchanges in each process is given by;
$Q_{\text{total}} = Q_{1\to2} + Q_{2\to3} + Q_{3\to1}$
$\Rightarrow Q_{\text{total}} = 60 \text{ J} - 40 \text{ J} + Q_{3\to1}$
$\Rightarrow Q_{\text{total}} = 20 \text{ J} \quad \ldots (2) \qquad [Q_{3\to1} = 0]$
The total work done in the cycle is:
$W_{\text{total}} = W_{1\to2} + W_{2\to3} + W_{3\to1}$
$W_{\text{total}} = 60 \text{ J} + W_{3\to1} \quad \ldots (3)$
From equations $(1), (2)$ and (3) we get;
$60 \text{ J} + W_{3\to1} = 20 \text{ J}$
$\Rightarrow 60 \text{ J} - U_{3\to1} = 20 \text{ J} \qquad [W_{3\to1} = -U_{3\to1}]$
$\Rightarrow U_{3\to1} = 60 \text{ J} - 20 \text{ J} = 40 \text{ J}$
Hence, option **(4)** is the correct answer.

Q2) correct option is c

Q3) **Explanation:**

Hint: Recall the degree of freedom for the gases.

Step: Identify the correct match.
A monatomic gas (e.g., helium, neon) has 3 translational degrees of freedom (movement along x, y, z axes).
A diatomic gas without vibration (e.g., oxygen, nitrogen at low temperatures) has 3 translational degrees of freedom and 2 rotational degrees of freedom (rotation around two axes). A diatomic gas without vibration has total 5 degree of freedom.
A diatomic gas with vibration (e.g., oxygen, nitrogen at high temperatures) has 3 translational degrees of freedom, 2 rotational degrees of freedom (rotation around two axes), and 2 vibrational degrees of freedom (contributing 2 due to kinetic and potential energy components). So, a diatomic gas with vibration has total 7 degree of freedom.
A triatomic gas (e.g., water) without vibration has 3 translational degrees of freedom and 3 rotational degrees of freedom (rotation around all three axes for nonlinear molecules). So, a triatomic gas without vibration has total 6 degree of freedom.
Therefore, the correct match is $\text{A-IV}, \text{B-III}, \text{C-I}, \text{D-II}.$
Hence, option **(1)** is the correct answer.

Q4) **Hint:** $PV^{\gamma}=cons\tan t$

Step 1: Use the first law of thermodynamics.

$\Delta Q=W+\Delta U$

$0=W+\Delta U$

$\Delta U=-W$

$\Delta U\ is\ +ve$

$T\uparrow$

Step 2: Use the ideal gas equation.

$PV=nRT$

$P=\frac{nRT}{V}$

$P\uparrow$

As volume is decreasing and temperature is increasing so P will increase.

Q5) **Explanation:**

Hint: $U\propto T$

Step: Find the correct P-V graph.

From the P-U graph, the pressure of the gas is directly proportional to the internal energy of the gas i.e., $P\propto U$.

For the ideal gas, the internal energy of the gas is directly proportional to the temperature of the gas i.e., $U\propto T$

$\Rightarrow P\propto T \quad \ldots(1)$

According to the ideal gas equation;

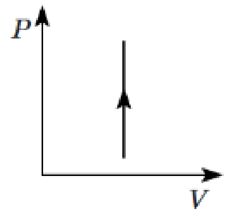
$PV=nRT$

$\Rightarrow P=\frac{nR}{V}T \quad \ldots(2)$

From the equation (1) and (2) we get;

If P increases as U (and hence T) increases, the volume V must remain constant for this direct proportionality to hold.

Therefore, the correct P-V graph is shown in the figure below;



Hence, option **(2)** is the correct answer.

Q6) **Explanation:**

Hint: Apply Newton's law of cooling.

Step 1: Use Newton's law of cooling for given situations.

$$\frac{20}{5} = K(70 - 20) \quad \ldots.(1)$$

also $\frac{20}{t} = K(50 - 20)\ldots.(2)$

Step 2: Find the time taken to cool from 60°C to 40°C.

After solving equation (1) & (2), we get;

$\Rightarrow t = \frac{25}{3}$ min

Hence, option (1) is the correct answer.

Q7) **Explanation:**

Hint: The gas will be at its highest temperature at the point where it has the highest pressure and volume.

Step: Find the point at which the gas reaches its highest temperature.

By using the ideal gas equation;

$PV = nRT$

$\Rightarrow T = \frac{PV}{nR}$

The temperature of the gas at point A is given by;

$\Rightarrow T_A = \frac{P_0V_0}{nR}$

The temperature of the gas at point B is given by;

$\Rightarrow T_B = \frac{3P_0V_0}{nR}$

The temperature of the gas at point C is given by;

$\Rightarrow T_C = \frac{9P_0V_0}{nR}$

The temperature of the gas at point D is given by;

$\Rightarrow T_D = \frac{8P_0V_0}{nR}$

The temperature of the gas at point E is given by;

$\Rightarrow T_D = \frac{3P_0V_0}{nR}$

Therefore, the gas reaches its highest temperature at point C.

Hence, option **(3)** is the correct answer.

Q8) **Explanation:**

Hint: $C_V = \frac{f}{2}R$

Step 1: Find the degree of freedom.
For each additional vibrational mode degree of freedom is increased by 2 so the new degree of freedom,
$f = 3 + 2 + 2 = 7$

Step 2: Find the molar-specific heat capacity of an isochoric process.
Volume is constant in an isochoric process.
$\Rightarrow C_V = \frac{f}{2} \times R = \frac{7}{2}R$
Hence, option **(3)** is the correct answer.

Q9) **Explanation:**

Hint: The molar volume of water at STP $= 22.4$ Litre

Step: Find the volume of water with a mass of 4.5 kg.
The number of moles of water is given by;
$n = \frac{4.5 \times 1000}{18}$
The volume of the water is given by;
$V =$ (no. of moles) (22.4 litre)
$V = \frac{\text{mass}}{\text{molar mass}} \left(22.4 \times 10^{-3}\,\mathrm{m}^3\right)$
$V = \frac{4.5 \times 1000}{18} \times 22.4 \times 10^{-3}$
$\Rightarrow V = 5.6\ \mathrm{m}^3$
Hence, option **(1)** is the correct answer.

Q10) **Explanation:**

Hint: $C_p - C_v = R$

Explanation: If a mixture of ideal gases has specific molar heat capacities C_p (at constant pressure) and C_v (at constant volume) then, $C_p - C_v = R$. Any mixture of ideal non-reacting gases will obey the ideal gas law: $pV = nRT$.
Therefore, both **Statement I** and **Statement II** are correct.
Hence, option **(2)** is the correct answer.

Q11) **Explanation:**

Hint: $C_P - C_V = R$

Step: Identify the correct match.

The molar specific heat of a monoatomic gas at constant volume is given by;

$C_V = \frac{fR}{2} = \frac{3R}{2} \qquad [f = 3]$

The molar specific heat of a monoatomic gas isobaric process (constant volume) is given by;

$C_P = \frac{3R}{2} + R = \frac{5R}{2}$

The molar specific heat of a diatomic gas in an isochoric process (constant volume) is given by;

$C_V = \frac{fR}{2} = \frac{5R}{2} \qquad [f = 5]$

The Bulk modulus of elasticity as a multiple of pressure for a monatomic gas undergoing an adiabatic process is given by; $B = \gamma P$

$\Rightarrow B = \frac{5}{3}P \qquad \left[\gamma = \frac{C_p}{C_V} = \frac{\frac{5R}{2}}{\frac{3R}{2}} = \frac{5}{3}\right]$

The Bulk modulus of elasticity as a multiple of pressure for a diatomic gas undergoing an isothermal process is given by;

$B = P$

Therefore, the correct match is $\text{A-IV, B-IV, C-III, D-I}$.

Hence, option **(1)** is the correct answer.

Q12) **Explanation:**

Hint: $v_{rms} \propto \sqrt{T}$

Step 1: Find the relation between the temperatures.

$PV = nRT$

$\Rightarrow T \propto PV$

$$\Rightarrow \frac{T_2}{T_1} = \frac{P_2V_2}{P_1V_1}$$

$$\Rightarrow \frac{T_2}{T_1} = \frac{4 \times 10^5 \times 1}{2 \times 10^5 \times 1}$$

$$\Rightarrow \frac{T_2}{T_1} = 2$$

Step 2: Obtain ratio of rms speeds.

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

$$\Rightarrow \frac{v_{2,rms}}{v_{1,rms}} = \sqrt{\frac{T_2}{T_1}}$$

$$\Rightarrow \frac{v_{2,rms}}{v_{1,rms}} = \sqrt{2}$$

Hence, option **(2)** is the correct answer.

Q13) **Explanation:**

Hint: $\eta = 1 - \frac{T_2}{T_1}$

Step: Find the ratio of the absolute temperature of the hot reservoir to that of the cold reservoir, i.e., $\frac{T_1}{T_2}$.
The efficiency of the Carnot engine is given by;
$\eta = 1 - \frac{T_2}{T_1}$
$\Rightarrow 25\% = 1 - \frac{T_2}{T_1}$
$\Rightarrow \frac{1}{4} = 1 - \frac{T_2}{T_1}$
$\Rightarrow \frac{T_2}{T_1} = 1 - \frac{1}{4}$
$\Rightarrow \frac{T_2}{T_1} = \frac{3}{4}$
$\Rightarrow \frac{T_1}{T_2} = \frac{4}{3}$
Therefore, the ratio of the absolute temperature of the hot reservoir to that of the cold reservoir is $\frac{4}{3}$.
Hence, option **(4)** is the correct answer.

Q14) **Hint:** $Q = \Delta U + \Delta W$

Step 1: Find the change in internal energy.

$$C_V = \frac{3R}{2}$$
$$\Delta U = nC_V \Delta T$$
$$= \frac{3}{2}(nR\Delta T)$$
$$Here,\ nR\Delta T = \Delta(PV)$$
$$= P_2V_2 - P_1V_1$$
$$= 3P_0V_0$$
$$\Delta U = 4.5\ P_0V_0$$

Step 2: Find the work done.

$$W = \frac{1}{2}(3P_0)\,V_0$$
$$= 1.5\ P_0V_0$$

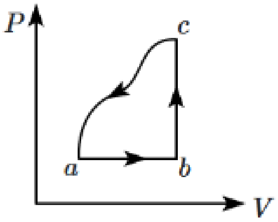
Step 3: Find the heat supplied.

$$\Delta Q = \Delta U + \Delta W$$
$$= 6P_0V_0$$

Q15) **Explanation:**

Hint: $\Delta Q = \Delta U + \Delta W$

Step: Find the work done by the gas along the path abc.
The ideal gas is taken through the cyclic process $abca$ is shown below;



Given: $\Delta U_{c\to a} = -180$ J , $\Delta U_{a\to c} = +180$ J (static function), $\Delta Q_{a\to b} = 250$ J , $\Delta Q_{b\to c} = 60$ J
According to the first law of thermodynamics,
$\Delta Q = \Delta U + \Delta W$
$\Rightarrow \Delta Q_{a\to c} = \Delta U_{a\to c} + \Delta W_{a\to c}$
$\Rightarrow \Delta Q_{a\to b} + \Delta Q_{b\to c} = 180 + \Delta W_{a\to c}$
$\Rightarrow 250 + 60 = 180 + \Delta W_{a\to c}$
$\Rightarrow \Delta W_{a\to c} = 250 + 60 - 180$
$\Rightarrow \Delta W_{a\to c} = 130$ J
Hence, option **(2)** is the correct answer.

Q16)

Solid Hollow

$\Delta Q = mc\Delta T$

$\Delta T = \frac{\Delta Q}{mc}$

For Hollow sphere, ΔT is more.

(as m is less for hollow sphere)

Now,

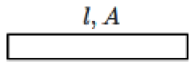
$\frac{\Delta V}{V} = \gamma \Delta T$

Therefore, the expansion in hollow is greater than the solid.

Q17) **Hint:** $H = \frac{KA\Delta T}{l}$

Step 1:

Find the thermal resistance of individual rods.

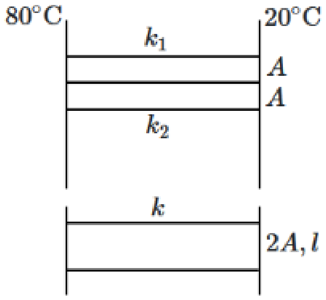


$R = \frac{l}{kA}$

$R_1 = \frac{l}{k_1 A} \quad R_2 = \frac{l}{k_2 A}$

Step 2:

Draw a diagram for the combination.



Step 3:

Find equivalent thermal resistance.

$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$

$\frac{k(2A)}{l} = \frac{k_1 A}{l} + \frac{k_2 A}{l}$

$\frac{k}{k_1} = \frac{1}{2}\left[1 + \frac{4}{3}\right] = \frac{7}{6}$

Q18) **Explanation:**

Hint: $\lambda = \frac{1}{\sqrt{2}\pi n d^2}$

Explanation: The mean free path of the gas is given by; $\lambda = \frac{1}{\sqrt{2}\pi n d^2}$

$\lambda \propto \frac{1}{n}$

If the number density of a gas in a vessel is increased then the mean free path decreases.

Hence, option **(1)** is the correct answer.

Q19) **Explanation:**

Hint: The rate of radiation, $H \propto T^4$

Step: Find the final rate of radiation.

As the rate of radiation, $H \propto T^4$ where the temperature is in Kelvin.

But the temperature is doubled in °C, the final temperature is less than doubled in Kelvin.

So the final rate of radiation will be less than 16-fold the initial rate of radiation.

Hence, option **(3)** is the correct answer.

Q20) **Explanation:**

Hint: Use the Mayor's formula.

Step: Find the molar-specific heats of an ideal gas at constant volume C_V.

$C_P - C_V = R$

Divide both sides by C_V we get;

$\frac{C_P}{C_V} - \frac{C_V}{C_V} = \frac{R}{C_V}$

$\Rightarrow \gamma - 1 = \frac{R}{C_V}$

$\Rightarrow C_V = \frac{R}{(\gamma - 1)}$

Hence, option **(1)** is the correct answer.

Q21) **Explanation:**

Hint: The rate of heat flow will be equal from both plates.

Step: Find the approximate temperature at the soldered junction.

we know that the rate of heat flow will be equal from both plates, i.e

$$\left(\frac{dQ}{dt}\right)_1 = \left(\frac{dQ}{dt}\right)_2 \quad \ldots(1)$$

$$K_1 A \frac{(100 - T)}{l} = K_2 A \frac{(T - 0)}{l} \quad \ldots(2)$$

putting the values in the equation we will get $48\,(100 - T) = 68\,(T)$

$$\Rightarrow T = \frac{1200}{29} = 41^\circ\text{C}$$

Hence, option (3) is the correct answer.

Q22) **Explanation:**

Hint: Use Newton's law of cooling.

Step: Find the steady temperature of the ball.

By applying Newton's law of cooling for the initial condition we get;

$\frac{dQ_1}{dt} = 30\ W = k\left(50^\circ\text{C} - 20^\circ\text{C}\right) \quad \ldots(1)$

By applying Newton's law of cooling for the final condition we get;

$\frac{dQ_2}{dt} = 20\ W = k\left(T^\circ\text{C} - 30^\circ\text{C}\right) \quad \ldots(2)$

By dividing equation (2) by (1) we get;

$\frac{2}{3} = \frac{T^\circ\text{C} - 30^\circ\text{C}}{50^\circ\text{C} - 20^\circ\text{C}}$

$\Rightarrow 20^\circ\text{C} = T^\circ\text{C} - 30^\circ\text{C}$

$\Rightarrow T^\circ = 50^\circ\text{C}$

Hence, option **(2)** is the correct answer.

Q23) **Explanation:**

Hint: $C_P - C_V = R$

Step: Find the molar-specific heat (at constant pressure) of an equimolar mixture.
C_P for monoatomic gas = $C_V + R = \frac{3}{2}R + R = \frac{5}{2}R$
C_P for diatomic gas = $C_V + R = \frac{5}{2}R + R = \frac{7}{2}R$
C_P of an equimolar mixture of the two will be:
$$C_P = \frac{\frac{5R}{2} + \frac{7R}{2}}{2}$$
$$= 3R$$
Hence, option (3) is the correct answer.

Q24) **Explanation:**

Hint: $dQ = ms\Delta T$

Step: Calculate the temperature of the furnace.
Heat supplied by the furnace = heat gained by cold water and calorimeter
$250 \times 0.1 \times (T - 50^\circ\text{C}) = (240 + 10) \times 1 \times (50^\circ\text{C} - 20^\circ\text{C})$ [10 g of water equivalent to mass of calorimeter]
$\Rightarrow 0.1 \times (T - 50^\circ\text{C}) = 30^\circ\text{C}$
$\Rightarrow T - 50^\circ\text{C} = 300^\circ\text{C}$
$\Rightarrow T = 350^\circ\text{C}$
Hence, option **(2)** is the correct answer.

Q25) **Explanation:**

Hint: $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

Step: Find the rise in the temperature of the gas.
The ideal gas equation is given by;
$PV = nRT$
For the constant pressure, $V \propto T$
$\Rightarrow \frac{T_1}{T_2} = \frac{V_1}{V_2} = \frac{1}{2}$
$\Rightarrow T_2 = 2T_1$
$\Rightarrow T_2 + 273 = 2 \times (27 + 273)$
$\Rightarrow T_2 + 273^\circ\text{C} = 600^\circ\text{C}$
$\Rightarrow T_2 = 327^\circ\text{C}$
The rise in the temperature is given by;
$\Delta T = T_2 - T_1 = 327^\circ\text{C} - 27^\circ\text{C} = 300^\circ\text{C}$
Hence, option **(3)** is the correct answer.

Q26) **Explanation:**

Hint: Recall the law of equipartition of energy.

Step: Find the variation of ω_{rms} with temperature.

The energy associated with each degree of freedom is given by: $\frac{1}{2}kT$. A diatomic molecule in a gas has two rotational degrees of freedom. Hence:

$\Rightarrow E_{rms} = 2.\frac{1}{2}kT = \frac{1}{2}I\omega_{rms}^2$

$\Rightarrow \omega_{rms} \propto T^{1/2}$

Hence, option **(1)** is the correct answer.

Q27) **Explanation:**

Hint: $K.E_{(avg)} = \frac{3}{2}kT$

Explanation: The average internal kinetic energy, due to the translational motion of a molecule in an ideal gas at absolute temperature T is given by;

$K.E_{(avg)} = \frac{3}{2}kT \Rightarrow K.E_{(avg)} \propto T$

Therefore, the average kinetic energy is directly proportional to the absolute temperature and independent of the mass of the molecule and the number of atoms in a molecule.

Hence, option **(1)** is the correct answer.

Q28) **Explanation:**

Hint: Change in internal energy is the same in both cases.

Step 1: Find the extra work done in the path AC.

$W_{AC} = \frac{1}{2} \times 210$ cc $\times 40$ kPa $= 4.2$ J $= 1$ cal

Step 2: Find the heat required in the path AC.

$\Rightarrow Q_{AC} = Q_{ABC} + W_{AC}$

$\Rightarrow Q_{AC} = 20$ cal $+ 1$ cal $= 21$ cal

Hence, option (3) is the correct answer.

Q29) **Explanation:**

Hint: $C_{V_{mix}} = \frac{C_{V_1} + C_{V_2}}{n_1 + n_2}$

Step: Find the ratio of specific heats (C_P/C_V) of the mixture.

The specific heat of the mixture at constant volume is given by;

$C_{V_{mix}} = \frac{C_{V_1} + C_{V_2}}{n_1 + n_2}$

$\Rightarrow C_{V_{mix}} = \frac{\frac{5R}{2} + \frac{3R}{2}}{2} = \frac{8R}{4} = 2R$

The specific heat of the mixture at constant pressure is given by;

$C_{P_{mix}} = \frac{C_{P_1} + C_{P_2}}{n_1 + n_2}$

$\Rightarrow C_{P_{mix}} = \frac{\frac{7R}{2} + \frac{5R}{2}}{2} = \frac{12R}{4} = 3R$

The ratio of specific heats $(C_{P_{mix}}/C_{V_{mix}})$ of the mixture is given by;

$\frac{C_{P_{mix}}}{C_{V_{mix}}} = \frac{3R}{2R} = \frac{3}{2}$

Hence, option **(2)** is the correct answer.

Q30) **Explanation:**

Hint: $PV^{\gamma}=$ constant

Step: Find the temperature of the gas.

For the adiabatic process, the equation of the state is given by; $PV^{\gamma}=$ constant $\quad ...(1)$

The ideal gas equation is given by; $PV=nRT \Rightarrow V=\frac{nRT}{P} \quad ...(2)$

From equations (1) and (2) we get;

$P^{1-\gamma}T^{\gamma}=$ constant

$\Rightarrow P^{(1-1.5)}T^{1.5}=$ constant $\Rightarrow P^{(-0.5)}T^{1.5}=$ constant

$\Rightarrow T^{1.5}=$ constant $P^{0.5}$

Taking log both side we get;

$1.5logT=log(\text{constant})+0.5logP$

After taking derivative both side we get;

$1.5\left(\frac{dT}{T}\right)=0+0.5\left(\frac{dP}{P}\right)$

$\Rightarrow \frac{dT}{T}=\frac{1}{3}\left(\frac{dP}{P}\right) \Rightarrow \frac{dT}{T}\times 100=\frac{1}{3}\left(\frac{dP}{P}\times 100\right)=\frac{1}{3}\%$

Therefore, the temperature of the gas increases by $\frac{1}{3}\%$

Hence, option (3) is the correct answer.

Q31) **Explanation:**

Hint: $Q=mL$

Step: Find the amount of ice needed to condense m gram of stream.

The heat released during condensation is given by;

$Q_{\text{condensation}}=m\times 540$ cal

The heat released during cooling is given by;

$Q_{\text{cooling}}=m\times 100$ cal

The total heat released by the steam is given by;

$Q_{total}=Q_{\text{condensation}}+Q_{\text{cooling}}=m(540+100)=640m$ cal

The heat absorbed by the ice is given by;

$Q_{ice}=m_{ice}\times 80$ cal

The heat released by the steam must equal the heat absorbed by the ice,

$Q_{ice}=Q_{total}$

$\Rightarrow m_{ice}\times 80=640m$

$\Rightarrow m_{ice}=\frac{640m}{80}=8m$ grams

Hence, option **(3)** is the correct answer.

Q32) **Explanation:**

Hint: $v_{rms} = \sqrt{\frac{3k_BT}{m}}$

Step: Find the average momentum of the gas molecules, in a direction parallel to a side of the vessel.
The root-mean-square speed (v_{rms}) of gas molecules, which represents the typical speed of molecules in the gas is given by;
$v_{rms} = \sqrt{\frac{3k_BT}{m}}$
While individual molecules may have non-zero velocities, the distribution of velocities ensures that the average momentum in any particular direction is zero because all momentum will cancel each other.
Therefore, the average momentum of the gas molecules is zero.
Hence, option (4) is the correct answer.

Q33) **Explanation:**

Hint: Recall the concept of thermodynamics variable (P, V, T)

Explanation: The equilibrium state of a thermodynamic system is defined by a set of conditions where the system is in a stable, unchanging state. These conditions include:
Pressure: A measure of the force exerted by the system per unit area. At equilibrium, the pressure within the system is constant and uniform.
Temperature: A measure of the average kinetic energy of the particles within the system. At equilibrium, the temperature throughout the system is constant and uniform.
Volume: The amount of space occupied by the system. At equilibrium, the volume remains constant unless external forces act upon the system.
Let's examine why the other options are incorrect:
Total heat: While heat transfer can occur during a process, the total heat content of a system is not a defining characteristic of its equilibrium state. Equilibrium is about the absence of net change, not the total amount of energy present.
Work done: Work is a process, not a state variable. It describes the energy transfer during a change in the system, not its equilibrium condition.In summary, pressure, temperature, and volume are the key state variables that define the equilibrium state of a thermodynamic system because they are constant and uniform under these conditions.
Hence, option **(4)** is the correct answer.

Q34) **Explanation:**

Hint: $W = P\Delta V$.

Step 1: Identify the area of the isobaric process.
Here to calculate the work done by the isobaric process we'll take the entire area.
Step 2: Find the work done during the isobaric process.
$W_{\mathrm{AB}} = P\Delta V$
$\Rightarrow W_{\mathrm{AB}} = (3 \times 10^2 - 0) \times (3 - 1)$
$\Rightarrow W_{\mathrm{AB}} = 600$ J
Hence, option **(4)** is the correct answer.

Q35) **Explanation:**

Hint: $R_{th} = \frac{L}{KA}$

Step: Find the ratio of the thermal current in BC to that in AD.
The heat current in the rod is given by;
$I = KA\frac{\Delta\theta}{L}$
Since the rods are identical, the thermal resistance of each rod is the same i.e., $R_{th} = \frac{L}{KA}$
The resistance involves paths of $AB + BC + CD$ and AD. Due to symmetry, the ratio of the thermal current in BC to that in AD is $\frac{1}{3}$.
Hence, option **(3)** is the correct answer.

Q36) **Explanation:**

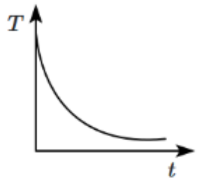
Hint: $Q = \frac{KA(T_1 - T_2)t}{L}$

Step: Find the thermal conductivity of the slab.
The heat flows through the slab in t s is given by;
$Q = \frac{KA(T_1 - T_2)t}{L}$
Substituting the known values we get;
$Q = \frac{K \times 0.36 \times (100 - 0) \times 3600}{0.1}$
$Q = \frac{K \times 0.36 \times 100 \times 3600}{0.1} \quad \ldots(1)$
So ice melted by this heat is given by;
$m_{\text{ice}} = \frac{Q}{L_f}$
or $Q = m_{\text{ice}} L_f$
$Q = 4.8 \times 3.36 \times 10^5 \quad \ldots(2)$
From (1) and (2), we get;
$\frac{K \times 0.36 \times (100 - 0) \times 3600}{0.1} = 4.8 \times 3.36 \times 10^5$
$K = \frac{4.8 \times 3.36 \times 10^5 \times 0.1}{0.36 \times 100 \times 3600}$
$\Rightarrow K = 1.24 \text{ J/m/s/}^\circ\text{C}$
Hence, option **(4)** is the correct answer.

Q37) **Explanation:**

Hint: Temperature decreases exponentially with time.

Step: Find the correct graph for the temperature (T) and time (t).
According to Newton's law of cooling; the loss of heat by radiation depends upon the nature of the surface of the body and the area of the exposed surface i.e.,
$-\frac{dQ}{dt} = k(T_2 - T_1)$
$\Rightarrow -ms\frac{dT_2}{dt} = k(T_2 - T_1) \qquad \left[\frac{dQ}{dt} = ms\frac{dT}{dt}\right]$
$\Rightarrow \frac{dT_2}{dt} = -\frac{k}{ms}(T_2 - T_1)$
$\Rightarrow \frac{dT_2}{T_2 - T_1} = -\frac{k}{ms}dt$
Integrating both sides we get;
$\Rightarrow \int \frac{dT_2}{T_2 - T_1} = \int -\frac{k}{ms}dt$
$\Rightarrow \text{Ln}(T_2 - T_1) = -\frac{k}{ms}t + c$
$\Rightarrow (T_2 - T_1) = e^{-\frac{k}{ms}t} + c$
$\Rightarrow T_2 = T_1 + e^{-\frac{k}{ms}t} + c$
Therefore, the correct graph of the temperature (T) and time (t) is shown in the figure below;



Hence, option **(2)** is the correct answer.

Q38) **Explanation:**

Hint: $Q = ms\Delta T$

Step: Identify the correct answer.
Heat lost by iron = Heat gained by (water + plastic cup)
$m_{\text{iron}} s_{\text{iron}} \Delta T_{\text{iron}} = m_{\text{water}} s_{\text{water}} \Delta T_{water} + W \Delta T_{cup}$
So, to find the specific heat of the iron block, the thermal conductivity of iron is not required.
Hence, option **(3)** is the correct answer.

Q39) **Explanation:**

Hint: Use Stefan's law.

Step: Find the temperature of the given black body.
$E = \sigma A T^4$
So, the rate of energy production is given by;
$Q = E \times 4\pi \mathrm{R}^2$
$Q = \sigma T^4 \times 4\pi R^2$
The temperature of the star is given by;
$\Rightarrow T = \left(\frac{Q}{4\pi\sigma R^2}\right)^{1/4}$
Hence, option **(4)** is the correct answer.

Q40) **Explanation:**

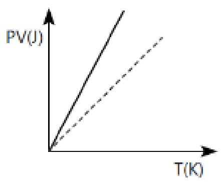
Hint: $\lambda_{mp} \propto \frac{1}{T}$.

Step: Find the relationship between v_{rms} and λ_{mp}.
According to the Wien's Displacement Law: $\lambda_{mp} T =$ constant
$\lambda_{mp} \propto \frac{1}{T} \ldots (1)$
$v_{rms} = \sqrt{\frac{3RT}{M}}$
$v_{rms}^2 \propto T \ldots (2)$
From (1) and (2)
$\lambda_{\text{mp}} v_{\text{rms}}^2 =$ constant
Hence, option (3) is the correct answer.

Q41) **Explanation:**

Hint: $PV = nRT$

Step: Identify the correct graph.
The slope of graph $PV = T$ is 1.
The ideal gas follows the equation:
$PV = RT$
The slope of the above graph is R, which is greater than 1.
Therefore the graph shown below is the correct representation.



Hence, option **(1)** is the correct answer.

Q42) **Explanation:**

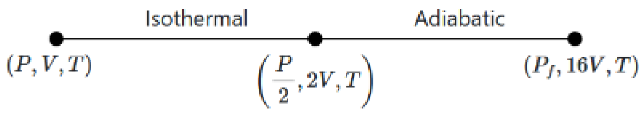
Hint: $Q = mL$

Step: Find the thermal power absorbed by the ice.
The thermal power absorbed by the ice is given by:
$P = \frac{Q}{t}$
$\Rightarrow P = \frac{mL}{t}$
$\Rightarrow P = (10 \text{ g/s}) \times (80 \text{ cal/g}) \quad \left[\because \frac{m}{t} = 10 \text{ ml/s} = 10 \text{ g/s}, L = 80 \text{ cal/g}\right]$
$\Rightarrow P = 800 \text{ cal/s}$
$\Rightarrow P = 800 \times 4.2 \text{ J/s} \quad [1 \text{ cal} = 4.2 \text{ J}]$
$\Rightarrow P = 3360 \text{ J/s}$
$\Rightarrow P = 3.36 \text{ kW}$
Hence, option **(1)** is the correct answer.

Q43) **Explanation:**

Hint: $PV^\gamma =$ constant

Step: Find the final pressure of the gas.
In the case of the isothermal process: $PV =$ constant
$\Rightarrow P_1V_1 = P_2V_2$
$\Rightarrow PV = P_2(2V)$
$\Rightarrow P_2 = \frac{P}{2}$



In adiabatic process: $PV^\gamma =$ constant
$\Rightarrow P_1V_1^\gamma = P_2V_2^\gamma$
$\Rightarrow P_2(2V)^\gamma = P_f(16V)^\gamma$
$\Rightarrow P_f = P_2\left(\frac{2V}{16V}\right)^\gamma$
$\Rightarrow P_f = \frac{P}{2} \times \left(\frac{1}{8}\right)^{\frac{5}{3}} = \frac{P}{2} \times \left(\frac{1}{2}\right)^5$
$\Rightarrow P_f = \frac{P}{64}$
Hence, option **(3)** is the correct answer.

Q44) **Explanation:**

Hint: Use the ideal gas equation to deduce the correct relation.

Step: Identify the equations representing the adiabatic process.
The adiabatic process of gas is represented by the equation;
$PV^\gamma =$ constant $\quad \ldots(1)$
$\Rightarrow TV^{\gamma-1} =$ constant $\quad \left[P = \frac{nRT}{V}\right] \quad \ldots(2)$
$\Rightarrow P^{1-\gamma}T^\gamma =$ constant $\quad \left[V = \frac{nRT}{P}\right] \quad \ldots(3)$
The equation $(1), (2)$ and (3) equations represent the adiabatic process.
Therefore, $P^\gamma T^{1-\gamma} =$ constant is not an equation for an adiabatic process.
Hence, option **(4)** is the correct answer.

Q45) **Explanation:**

Hint: $P = \frac{nRT}{V}$

Step: Find the internal energy per unit volume.
The internal energy of a monoatomic gas is given by:
$U = \frac{3}{2}nRT$
$\Rightarrow \frac{U}{V} = \frac{3}{2}(\frac{nRT}{V})$
$\Rightarrow u = \frac{3}{2}p$
$\Rightarrow p = \frac{2}{3}u$
Hence, option (1) is the correct answer.

Q46) **Explanation:**

Hint: Electron-releasing substituents activate the ring for electrophilic substitution, also considering steric factors.

Q47) Hint: The second reaction is an aromatization reaction.

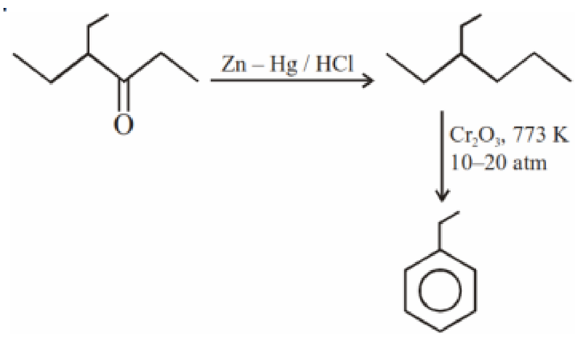
In the first reaction, reduction of the ketone group takes place and an alkane is obtained as a product.

In the second reaction, an aromatization of alkane takes place and benzene is obtained as a product.

n-Alkanes having six or more carbon atoms on heating to 773K at 10-20 atmospheric pressure in the presence of oxides of

vanadium, molybdenum or chromium supported over alumina get hydrogenated and cyclised to benzene and its homologues.

This reaction is known as aromatization or reforming. The reactions are as follows:



Q48) **Explanation:**

Hint: Wurtz reaction

Only methyl bromide can give methane and ethane as a product in two different conditions. When methyl bromide undergoes reduction with zinc and dilute hydrochloric acid then methane is obtained as a product.

The reaction is as follows:

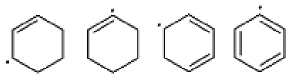
$$\underset{\text{Chloromethane}}{CH_3-Cl} + H_2 \xrightarrow{Zn,\ H^+} \underset{\text{Methane}}{CH_4} + HCl$$

Ethane is also formed from methyl bromide. When 2 moles of methyl bromide react with Na in the presence of diethyl ether then ethane is obtained as a product. The reaction is known as the Wurtz reaction.

The reaction is as follows:

$$\underset{\text{Bromomethane}}{CH_3Br} + 2Na + BrCH_3 \xrightarrow{\text{dry ether}} \underset{\text{Ethane}}{CH_3-CH_3} + 2NaBr$$

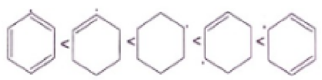
Q49) **Explanation:**



The image shows a set of organic compounds, each containing a radical (indicated by the dot). These radicals are located in different positions in benzene or its derivatives, affecting the stability of each compound.

To explain the stability order, let's consider the nature of the radicals and the resonance stability of the compounds:

1. **Cyclohexyl Radical (first structure)**: This is an alkyl radical without any resonance stabilization. It is the least stable because there is no delocalization of the unpaired electron.
2. **Benzyl Radical (second structure)**: The radical is on a carbon atom adjacent to a benzene ring (benzylic position). The radical is stabilized by resonance with the benzene ring, making it more stable than the cyclohexyl radical.
3. **Allyl Radical (third structure)**: This radical is at an allylic position relative to the double bond. The unpaired electron can delocalize over the double bond, making it moderately stable.
4. **Phenyl Radical (fourth structure)**: The radical is directly attached to a benzene ring. However, it cannot participate in the resonance of the benzene ring, making it less stable than the allyl radical but more stable than the cyclohexyl radical.
5. **Cyclohexadienyl Radical (fifth structure)**: The radical is part of a conjugated diene system. This allows the unpaired electron to be delocalized over multiple carbon atoms, making this the most stable radical in the series.

(i) 

Hence, **option 1** is the correct choice.

Q50) Explanation:

Step 1

In the Dumas method, a known quantity of nitrogen-containing organic compounds is heated strongly with an excess of copper oxide in an atmosphere of carbon dioxide to produce free nitrogen in addition to carbon dioxide and water.

Step 2

The chemical equation involved in the process can be represented as

$$CxHyNz + (2x + y/2)CuO \rightarrow xCO_2 + y/2H_2O + z/2N_2 + (2x + y$$

The traces of nitrogen oxides can also be produced in the reaction, which can be reduced to dinitrogen by passing the gaseous mixture over a heated copper gauge. The dinitrogen produced is collected over an aqueous solution of potassium hydroxide. The volume of nitrogen produced is then measured at room temperature and atmospheric pressure.

Step 3

On the other hand, in Kjeldahl's method, a known quantity of nitrogen-containing organic compounds is heated with concentrated sulphuric acid. The nitrogen present in the compound is quantitatively converted into ammonium sulphate. It is then distilled with excess sodium hydroxide. The ammonia evolved during this process is passed into a known volume of H_2SO_4.

Step 4

The chemical equations involved in the process are

$$Organic \quad compound \xrightarrow{Conc.H_2SO_4} (NH_4)_2 SO_4$$
$$(NH_4)_2 SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2NH_3 + 2H_2O$$
$$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2 SO_4$$

The acid that is left unused is estimated by volumetric analysis (titrating it against a standard alkali) and the amount of ammonia produced can be determined. Thus, the percentage of nitrogen in the compound can be estimated. This method cannot be applied to the compounds, in which nitrogen is present in a ring structure, and is also not applicable to compounds containing nitro and azo groups.

Q51) **Explanation:**

Molar mass of $Mg_2P_2O_7$ (magnesium pyrophosphate):

The formula is $Mg_2P_2O_7$.

Molar mass = 2(24.31) + 2(30.97) + 7(16) = 48.62 + 61.94 + 112 = **222.56 g/mol**

Mass of phosphorus in $Mg_2P_2O_7$

There are 2 phosphorus atoms in $Mg_2P_2O_7$, so the total mass of phosphorus in one mole of $Mg_2P_2O_7$ is

2 × 30.97 g = **61.94 g** of phosphorus in 222.56 g of $Mg_2P_2O_7$.

Using the ratio of mass:

From 0.555 g of $Mg_2P_2O_7$, the mass of phosphorus can be calculated using the ratio:

$$\text{Mass of } \mathbf{P} = \frac{\mathbf{61.94}}{\mathbf{222.56}} \times \mathbf{0.555\ g} = \mathbf{0.1545\ g}$$

Percentage of phosphorus in the organic compound:

The mass of the organic compound is 0.40 g.

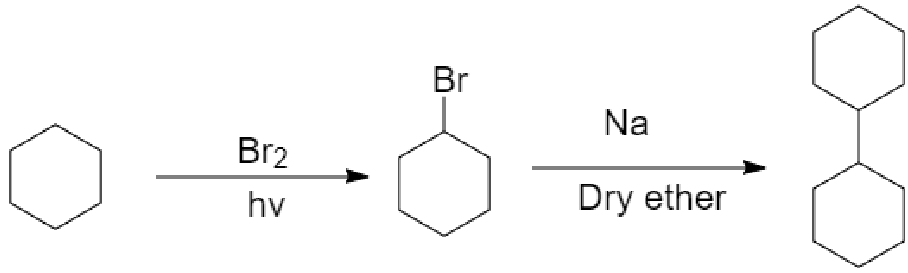
The percentage of phosphorus is:

$$\text{Percentage of } \mathbf{P} = \frac{\mathbf{0.1545\ g}}{\mathbf{0.40\ g}} \times \mathbf{100} = \mathbf{38.625\%} \approx \mathbf{38.75\%}$$

Q52) Hint: The second reaction is the Wurtz reaction

When cyclohexane reacts with Br_2 in the presence of light then a free radical substitution reaction occurs and bromocyclohexane is obtained as a product. In the next step, two moles of bromocyclohexane react with each other and a Wurtz reaction occurs.

The reaction is as follows:



Q53) **Explanation:**

Hint: In TLC, the separation occurs through absorption.

Explanation:

Thin-layer chromatography (TLC) is a chromatography technique that separates components in non-volatile mixtures. The principle of TLC is based on the separation of mixture components between a fixed stationary phase and a liquid mobile phase. The separation occurs through adsorption.

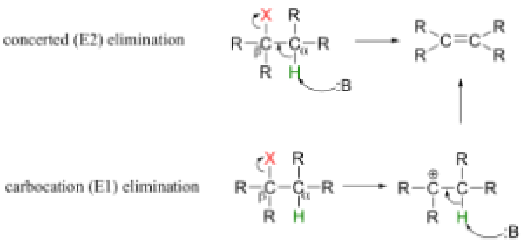
The process begins by moving the mobile phase over the stationary phase's surface. During this movement, the higher-affinity compounds gain less speed as compared to the lower-affinity compounds. This results in their separation.

Silica gel, or alumina, is coated on a flat support (usually glass or plastic) to form the stationary phase.

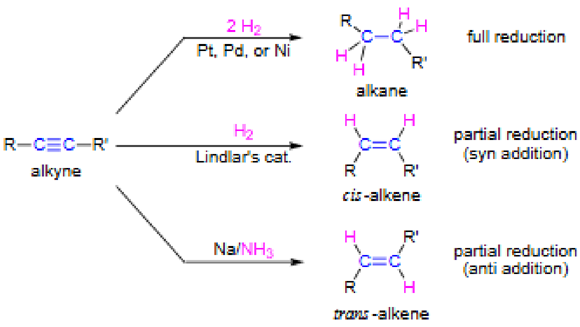
Hence, **option 3** is the correct choice.

Q54) **Explanation:**

(i) Alkenes are generally prepared through β elimination reactions, in which two atoms on adjacent carbon atoms are removed, resulting in the formation of a double bond. Preparations include the dehydration of alcohols, the dehydrohalogenation of alkyl halides, and the dehalogenation of alkanes.



(ii) The addition of alkynes with hydrogens (hydrogenation) forms alkenes. Alkynes react with dihydrogen in the presence of catalysts such as Pt/Pd/Ni to form alkenes.



Hence, alkenes can be prepared from addition as well as elimination reactions. Therefore, **option 4** is the correct answer.

Q55) **Explanation:**

Hint: NaSCN is formed.

Explanation:

Sodium combines with the carbon and the nitrogen present in the compound to form the cyanide. If sulphur is also present, sodium thiocyanate, NaSCN will also be formed.

$Na\ +\ C\ +\ N\ \rightarrow\ NaCN$ (If only nitrogen is present)

$Na\ + S\ +\ C\ +\ N\ \rightarrow\ NaSCN$ (If nitrogen and sulphur both are present)

Thiocyanate is a pseudohalide anion obtained by deprotonation of the thiol group of thiocyanic acid.

Hence, **option 2** is the correct answer.

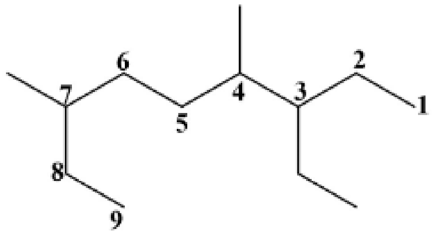
Q56) **Hint:** The parent chain will have highest number of carbon atoms.

Explanation:

(1) Here, the longest chain have nine atoms.

(2) Three substituents are present, one ethyl and two methyl groups. The numbering will start from the side where all the substituents get the lowest number.

(3) The substituents are named in an alphabetical order. "di" prefix will be used for two methyl groups. So, the correct name of the molecule is : 3-ethyl-4, 7-dimethynonane.



Hence, option 2 is the correct answer.

Q57) **Explanation:**

The whole reaction takes place in the given three steps.

Step 1: Initiation:

The reaction begins with the homolytic cleavage of Cl – Cl bond as:

$$Cl - Cl \xrightarrow{h\nu} Cl\cdot + Cl\cdot$$

Step 2: Propagation:

In the second step, chlorine-free radicals attack methane molecules and break down the C–H bond to generate methyl radicals as:

$$CH_4 \ + \ Cl \xrightarrow{h\nu} {}^{\cdot}CH_3 + HCl$$

These methyl radicals react with other chlorine-free radicals to form methyl chloride along with the liberation of a chlorine-free radical.

$${}^{\cdot}CH_3 + Cl - Cl \xrightarrow{h\nu} CH_3 - Cl + Cl\cdot$$

Hence, methyl-free radicals and chlorine-free radicals set up a chain reaction. While HCl and CH_3Cl are the major products formed, other higher halogenated compounds are also formed as:

$${}^{\cdot}CH_3 + Cl - Cl \xrightarrow{h\nu} {}^{\cdot}CH_2 - Cl + HCl$$

$${}^{\cdot}CH_2 - Cl + Cl - Cl \xrightarrow{h\nu} {}^{\cdot}Cl + CH_2Cl_2$$

Step 3: Termination:

The formation of ethane is a result of the termination of chain reactions taking place as a result of the consumption of reactants as:

$${}^{\cdot}Cl + {}^{\cdot}Cl \xrightarrow{h\nu} Cl - Cl$$

$${}^{\cdot}CH_3 + {}^{\cdot}CH_3 \xrightarrow{h\nu} H_3C - CH_3$$

Hence, by this process, ethane is obtained as a by-product of the chlorination of methane. Thus, **option 3** is the correct answer.

Q58) **Hint:** Include both ethers and alcohols.

Explanation:

(i) The degree of unsaturation in $C_4H_{10}O = \frac{2\times(C)+2-H}{2} = \frac{2\times4+2-10}{2} = 0$

So, it does not have any unsaturation. It is a saturated compound.

(ii) For the given formula, four different alcohols and three ethers are possible. Among all the isomers, butan-2-ol has one chiral carbon atom. So, two stereoisomers will be possible for butan-2-ol.

The total number of isomers for $C_4H_{10}O$ = 7 + 1 = 8

The possible isomers are shown as follows (excluding stereoisomers) :

The number of isomers in C4H10O are:

Hence, **option 4** is the correct choice.

Q59) correct option is a

Q60) Hint: More electronegative is the element, the more its -I effect.

The -I effect depends on the electronegativity of the atom or group of atoms directly attached to the carbon chain.

In the first option, −I < −Br < −Cl < −F as the electronegativity increases, the -I effect also increases. Hence, the first order is correct.

In the second option, the given order is correct because oxygen containing a positive charge is more electronegative than nitrogen-containing a positive charge. Oxygen is more electronegative than nitrogen. Hence, $-\overset{+}{N}R_3 < -\overset{+}{O}R_2$ is correct.

In the third option, the given order is wrong because nitrogen-containing a positive charge is more electronegative than -OR and -F. Hence, the correct order is - OR < - F < $-\overset{+}{N}R_3$

In the fourth, option, the given order is correct. The oxygen atom is more electronegative than sulphur. Hence, -SR has less -I effect than -OR.

Q61) Distillation under reduced pressure is **used to purify a liquid that has a tendency to decompose when heated to a high temperature**. Under the conditions of reduced pressure, the liquid will boil at a temperature lower than its boiling point, and as a result, the liquids will not degrade as they would otherwise.

Q62) **Explanation:**

Hint: The inductive effect is the electron displacement through sigma bonds.

Question No: **7** from Live Session Link Below

Q63) Explanation:

Hint: If molecule contains chiral center than molecule is optically active.

$$\begin{array}{c}
\begin{array}{ccccccccc} & & & & CH_3 & & & & \\ & & & & | & & & & \\ CH_3 & - & CH_2 & - & CH & - & CH_2 & - & CH_3 + Cl_2 \end{array} \\
\qquad\qquad\qquad (1\ Mole) \\
\downarrow
\end{array}$$

$$\begin{array}{ccccccccccccccccc}
 & & & & CH_3 & & & & & & & & CH_3 & & & & \\
 & & & & | & & & & & & & & | & & & & \\
CH_2 & - & CH_2 & - & CH & - & CH_2 & - & CH_3 + CH_3 & - & CH & - & CH & - & CH_2 & - & CH_3 \\
| & & & & & & & & & & & & & & & & \\
Cl & & & & & & & & & & & & & & & &
\end{array}$$

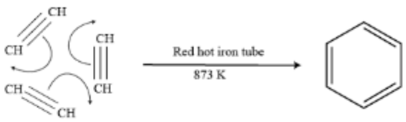
$$\begin{array}{ccccccccccccccccc}
 & & & & CH_3 & & & & & & & & CH - Cl & & & & \\
 & & & & | & & & & & & & & | & & & & \\
CH_3 & - & CH_2 & - & C & - & CH_2 & - & CH_3 + CH_3 & - & CH_2 & - & CH & - & CH_2 & - & CH_3
\end{array}$$

Q65) Explanation:

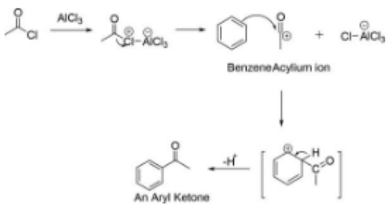
Hint: Y is benzene.

Explanation:

Step 1: Ethyne (the alkyne with the lowest molecular weight) passes through a red hot iron tube in the presence of Fe or Cu and undergoes cyclic polymerization to produce benzene.



Step 2: The benzene reacts with acetyl chloride in the presence of anhydrous aluminium chloride to form acetophenone. The reaction is a Friedel-Craft reaction in which an acyl group is attached to the aromatic ring.



So, "Z" will be acetophenone, which has a maximum number of 17 atoms in it's structure.

So, **option 4** is the correct choice.

Q66) Hint: Electron donating group increases the reactivity of the benzene ring towards EASR.

Step 1:

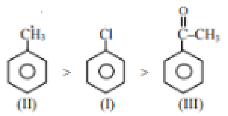
As the electron-donating group is attached to the benzene ring the electron density in the benzene ring increases and the reactivity of the benzene ring toward EASR increases. But in the case of the electron-withdrawing group reactivity of the benzene ring toward EASR decreases.

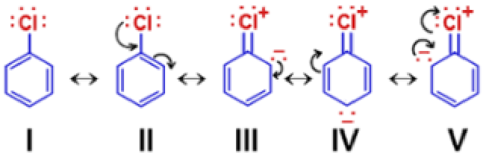
Step 2:

A CH_3 group is an electron-donating group hence it activates the benzene ring toward EASR. The $COCH_3$ group is a strong electron-withdrawing group it withdraws electrons by -R effect and deactivates the benzene ring towards EASR.

The chlorobenzene has intermediate reactivity because it shows a strong -I effect and weak +R effect. Hence, Cl is moderately deactivating.

Hence the reactivity order is as follows:

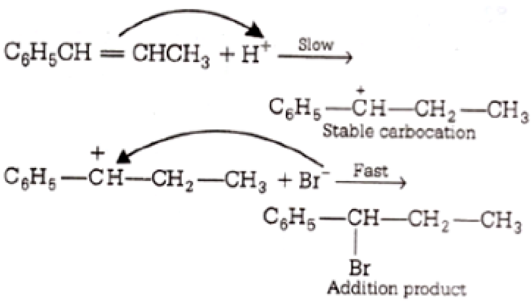




Q67) Explanation:

Hint: Benzylic carbocation is highly stable

It is an example of electrophilic addition reaction where proton addition takes place across the double bond resulting in more stable carbocation formation and subsequent addition of nucleophile to form addition product as shown below.



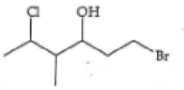
Q68) Explanation:

4. Boiling points of alkanes increase as the number of carbon atom increase or molecular mass increase. Alkenes have high boiling point, so $CH_3\,CH_2\,CH_2\,CH_3$ has the lowest boiling point.

Q69) Hint: The position of OH is third position

Step 1:

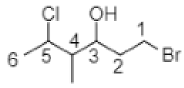
The structure of the given compound is as follows:



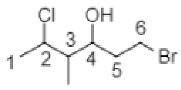
The longest carbon chain contains 6 carbon atoms. The numbering should be done in such a way that OH group containing carbon gets the lowest position.

Step 2:

The structure of a compound with numbering is as follows:



and



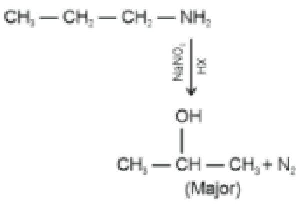
Left to right OH gets the third position and right to left it gets the fourth position. Hence, left to right numbering is followed.

In the first position, Br is attached, at the fourth carbon methyl group is present, and at the 5th carbon, Cl is present. The IUPAC name of the compound is as follows:

1-bromo-5-chloro-4-methylhexan-3-ol

Q70) **Explanation:**

Hint: The addition reaction of HBr to unsymmetrical alkenes follows Markovnikov Rule



Q71) **Explanation:**

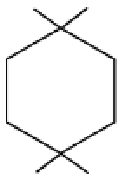
(i) A primary (1°) carbon atom is a carbon atom that is attached to only one other carbon atom, which means it is at the end of a carbon chain.

(ii) A secondary (2°) carbon atom is a carbon atom that is bonded to two other carbon atoms and has sp^3 hybridization.

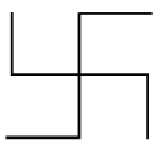
(iii) A tertiary (3°) carbon atom is a carbon atom that is attached to three other carbon atoms.

(iv) In a quaternary (4°) carbon atom, all four valencies of the carbon atom are satisfied by the other four carbon atoms.

Among the given compounds, the compounds that only $1^{\circ}, 2^{\circ}$ and 4° carbon atoms are:



, and



Hence, **option 3** is the correct answer.

Q72) **Explanation:**

Isomers are molecules with the same molecular formula but different structural or spatial arrangements of atoms. Isomers may have different physical properties, such as boiling point, melting point, and chemical reactivity.

In (1) the molecular formulas of the compounds are C_2H_6 *and* C_3H_8. Hence, both are not isomers.

In (2) the molecular formulas of the compounds are C_2H_4O *and* C_2H_6O. Hence, both are not isomers.

But in (3) the molecular formula of the compounds is C_4H_{10}. The structure is different. Hence, both are isomers.

Thus, **option 3** is the correct answer.

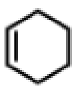
Q73) **Explanation:**

(i) An acyclic compound is a compound that does not have a cyclic structure.

(ii) Aromatic compounds are a group of compounds that have a cyclic conjugated structure and contain [4n+2]π electrons, making them particularly stable.

(iii) Antiaromatic compounds are cyclic molecules with a π electron system that have higher energy and are less stable than aromatic compounds. They contain 4n delocalized (π or lone pair) electrons and are flat.

(iv) an alicyclic compound contains one or more all-carbon rings which may be either saturated or unsaturated, but do not have aromatic character.

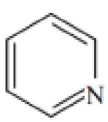


Given compound has a ring which is not aromatic or antiaromatic. Hence, **option 1** is the correct choice.

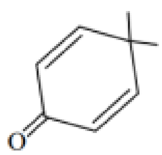
Q74) Hint: Non-aromatic molecule is the least resonance stabilized

If compound is aromatic than it is the most resonance stabilized.

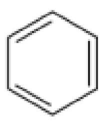
The compound given in option two is a non aromatic compound because ring contains sp^3 carbon atoms. Other compound are aromatic in nature. Hence, option two is the correct answer.



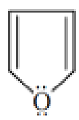
Aromatic



Non Aromatic



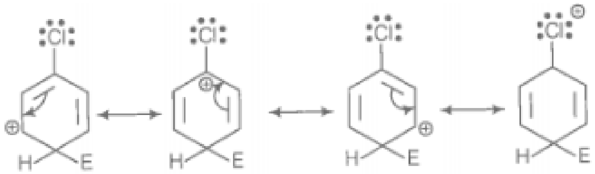
Aromatic



Aromatic

Q75) **Explanation:**

For an electrophilic substitution reaction, the presence of halogen atom in the benzene ring deactivates the ring by inductive effect and increases the charge density at ortho and para position relative to meta position by resonance.
When chlorine is attached to benzene ring, chlorine being more electronegative pulls the electron i.e., -I-effect. The electron cloud of benzene is less dense. Chlorine makes aryl halide, moderately deactivating group. But due to resonance the electron density on ortho and para position is greater than in meta position.



The last structure contributes more to the orientation and hence halogen are o- and p-directors.

Q76) Hint: Controlled oxidation reactions

Alkanes on heating with a regulated supply of dioxygen or air at high pressure and in the presence of suitable catalysts give a variety of oxidation products.

The reaction is as follows:

$$CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3} + HCHO + H_2O$$

Ordinarily, alkanes resist oxidation but alkanes having tertiary H atoms can be oxidized to corresponding alcohols by potassium permanganate. The reaction is as follows:

$$(CH_3)_3CH \xrightarrow[\text{Oxidation}]{KMnO_4} (CH_3)_3COH$$

Q78) **Explanation:**

Hint: Common stationary phases are silica gel and alumina.

Explanation:

TLC is based on the principle of separation through adsorption. The separation relies on the relative empathy of compounds towards the mobile phase and stationary phase.

The stationary phase is a thin adsorbent material layer, usually silica gel or aluminium oxide, coated onto an inert plate surface, typically glass, plastic, or aluminium.

The sample is spotted onto one end of the TLC plate and placed vertically into a closed chamber with an organic solvent (mobile phase).

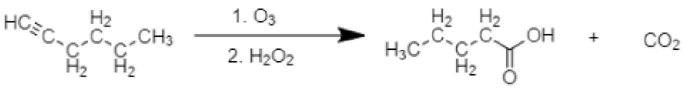
So, **option 3** is the correct choice.

Q79) Hint: Alkyne must contains a chiral center

C_6H_{10} is an alkyne. The alkyne is a terminal alkyne because C_6H_{10} reacts with sodaamide(base). Terminal alkyne has acidic hydrogen.

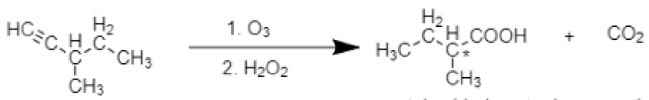
Hence, 3-Hexyne cannot be the answer because it is a non-terminal alkyne.

The ozonolysis reaction of other given alkyne is as follows:



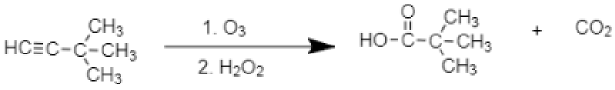
1-Hexyne

does not contains any chiral center hence optically inactive



contain chiral center hence optically active

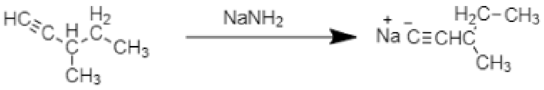
3-Methyl-1-pentyne



3,3-dimethyl-1-butyne

Hence, 3-methyl-1-pentyne gives chiral carboxylic acid on ozonolysis. It also contains a terminal triple bond hence it will react with sodamide.

The reaction is as follows:



3-Methyl-1-pentyne

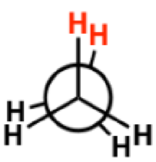
Q80) **Explanation:**

Hint: A dihedral angle or torsional angle (symbol: θ) is the angle between two bonds originating from different atoms in a Newman projection.

Explanation:

Eclipsed conformation of ethane is least stable because the dihedral angle is least in the eclipsed conformation. In eclipse conformation, the dihedral bond angle is zero.

0°



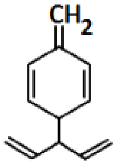
eclipsed

Q81) **Explanation:**

Hydrogenation is an addition reaction that occurs when hydrogen atoms are added to the carbon-carbon double bond of an alkene. The hydrogenation process breaks the pi bond in the double bond and replaces it with hydrogen atoms.

Hydrogenation of alkene takes place at $150\ ^{\circ}C$, whereas ring opening by hydrogenation takes place at $300\ ^{\circ}C$ (needs a different catalyst Ni/SiO_2 catalyst).

In the given compound five double bonds are present. So, five moles of hydrogen are required for complete hydrogenation of one mole of given compound.



Thus, **option 3** is the correct answer.

Q82) Explanation:

Hint: Resonance stabilized free radical is more stable than radical stabilized by inductive effect or hyperconjugation.

Explanation:

Because of the presence of the benzene ring, the benzylic radical has a total of five resonance contributors.

According to the resonance effect, the more resonance contributors available, the better the electron density will be dispersed, and the more stable the species will be.

With increase in number of benzene rings the stability of radical increases.

Thus, the correct order of stability will be : $(C_6H_5)_3\dot{C}$ > $(C_6H_5)_2\dot{C}\,CH_3$ > $C_6H_5\dot{C}(CH_3)_2$ > $(CH_3)_2\dot{C}H$

Hence, option 1 is the correct answer.

Q83)

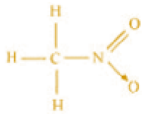
Nitrogen:	$Fe_4\left[Fe(CN)_6\right]_3$ Prussian Blue
Sulphur:	$[Fe\,(SCN)]^{2+}$ *or* $Fe(SCN)_3$
Phosphorous:	$(NH_4)_3PO_4.12MoO_3$
Halogen:	AgCl; AgBr; AgI

Q84) Explanation:
Step 1:
Alkanes experience inter-molecular Van der Waals forces. The stronger the force, the greater will be the boiling point of the alkane.
Step 2:
As branching increases, the surface area of the molecule decreases which results in a small area of contact. As a result, the Van der Waals force also decreases which can be overcome at a relatively lower temperature. Hence, the boiling point of an alkane chain decreases with an increase in branching.

Q85) Hint: A single bond is a sigma bond while a multiple bond is composed of one sigma bond together with pi or other bonds. A double bond has one sigma plus one pi bond, and a triple bond has one sigma plus two pi bonds.

Explanation:

Step 1: Below is the structure of CH_3NO_2



Step 2:

Count the sigma, and Pi bond in the above compound structure

There are three C–H sigma (σ_{C-H}) bonds, one C–N sigma (σ_{C-N}) bond, one N–O sigma (σ_{N-O}) bond, and one N=O pi (π_{N-O}) bond in the given compound.

Q86) Step 1: Chlorination ($Cl_2/FeCl_3$)

- NO_2 is a strong deactivating and meta-directing group.
- C_2H_5 (Ethyl) is an activating and ortho/para-directing group.
- However, NO_2 being a strong deactivator dominates, and chlorination occurs meta to NO_2.

  Intermediate: 4-Nitro-3-chloroethylbenzene

  Step 2: Reduction (Sn/HCl)

- The NO_2 group is reduced to NH_2, forming 4-Amino-3-chloroethylbenzene.

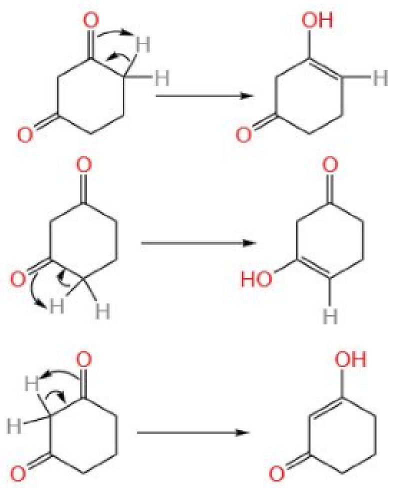
- Step 3: Diazotization ($NaNO_2$/HCl, 273–278 K)
- The NH_2 group is converted into a diazonium salt ($-N_2^+$).
- Intermediate: 4-Diazonium-3-chloroethylbenzene

- Step 4: Substitution with KI
- The $-N_2^+$ group is replaced by I, forming 4-Iodo-3-chloroethylbenzene.
- Final product identification:
- C_2H_5 at para position
- Cl at meta position
- I at para position to ethyl

Q87) **Explanation:**

Tautomerization is a net process by which protons are transferred from one site to another by a series of steps in which the solvent is an intermediary.

For the given molecule, following enol forms are possible.



It can't form a structure given in option 3. Thus, option 3 is the correct answer.

Q88) **Explanation:**

Hint: Count the Chiral carbons in the product.

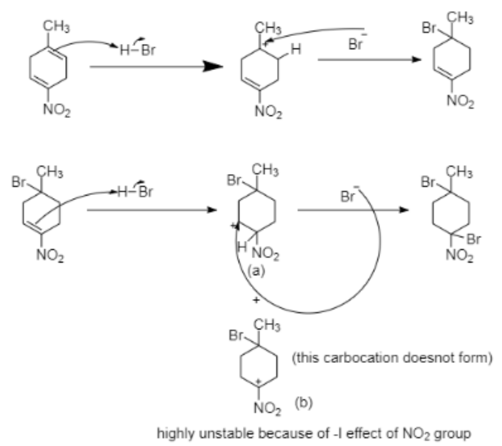
$(CH_3)_2C{=}CHCH_3 \rightarrow H2(CH_3)_2CH\text{-}CH_2CH_3$

the reduced product does not contain a chiral carbon atom and hence is optically inactive.

Q89) **Explanation:**

When alkene reacts with HBr then an alkyl bromide is obtained as a product. In the given reactant, two double bonds are present and 2 moles of HBr are given. Hence, both double bonds will react with HBr. The reaction is as follows:

The mechanism of the reaction is as follows:



Thus, the correct answer is **option 4.**

Q90)

$$\%\text{ of chlorine } = \frac{35.5}{143.5} \times \frac{\text{wt. }\%\text{ of chlorine}}{\text{wt. of organic compound}} \times 100$$

$$\%\text{ of Bromine } = \frac{80}{188} \times \frac{\text{wt. }\%\text{ of bromine}}{\text{wt. of organic compound}} \times 100$$

$$\%\text{ of Iodine } = \frac{127}{235} \times \frac{\text{wt. }\%\text{ of Iodine}}{\text{wt. organic compound}} \times 100$$

Mitoprep

Sapno Ki Udaan